

RESCUE JUNIOR - Bill of Materials (BOM)

Team name: Angora Goasbock

Every Line/Maze team has to submit a bill of materials for their robot

Instructions:

*Enter Team name above
*All costs need to be in local currency and their approximate conversion into dollars.
*All components worth less can be summarized in one line quantity 1 and their total cost (e.g. Screws)
*In the Local Currency Name column, the team must enter the name of the local currency, if it is different from US dollars.
*In the software sheet, in the Author column, the name of the designer, writer, or company that owns the software or library should be entered. If the team created their own algorithm, dataset, or AI model, they can include their names as authors.
*On the hardware sheet, in the "Author" column, enter the name of the company, engineer, or part manufacturer. If the team made a custom part, such as the robot housing, the team should include their names as the authors.

IMPORTANT The "Hardware" sheet is not required for simulation

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Name of the Local Currency	Total Cost Local Currency	Total Cost (\$) U.S.A. Dollars
Brazilian Real	R\$11 824,09	\$2 094,00

#	Component	Part name	Autor	Source	Quantity	Unit cost (Local)	Unit cost (USA Dollars)	Total Cost (Local)	Total Cost (USA Dollars)
1	PCB with electronic components	Custom made	Name of team members/Jlcpcb	jlcpcb.com	1	R\$338,80	\$60,00	R\$338,80	\$60,00
2	Wheels	Nexus Robot Mecanum Wheel	Nexus Robot	robotshop.com	4	R\$112,93	\$20,00	R\$451,72	\$80,00
3	Sheet metall parts for frame	Custom made		Local workshop	1	R\$1 129,34	\$200,00	R\$1 129,34	\$200,00
4	Drive motors	Maxon ECX TORQUE 22 XL Ø22 mm	Maxon Group	Maxon Motors	4	R\$1 694,01	\$300,00	R\$6 776,04	\$1 200,00
5	Robots shell	Custom made	Name of team members	3D Printer	1	R\$112,93	\$20,00	R\$112,93	\$20,00
6	Robots costume	Custom made	Name of team members	Self made	1	R\$282,34	\$50,00	R\$282,34	\$50,00
7	Microcontroller	Arduino Mega 2560	Arduino	conrad.de	2	R\$237,16	\$42,00	R\$474,32	\$84,00
8	Connecting parts	Screws, nuts, washers, ...		Local tool store	1	R\$225,87	\$40,00	R\$225,87	\$40,00
9	Motors for arm	NEMA-17-01 NEMA-17-01 0.4 Nm 1.68 A	Conrad Electronic	conrad.de	2	R\$118,58	\$21,00	R\$237,16	\$42,00
10	Stepper motor controllers	Pololu Tic T825	Electronics	conrad.de	2	R\$508,20	\$90,00	R\$1 016,40	\$180,00
11	LEDs	Various colours and sizes	Farnell An Avnet Company	farnell.com	20	R\$2,82	\$0,50	R\$56,40	\$10,00
12	Battery	11.1 V 3700 mAh	Farnell An Avnet Company	farnell.com	3	R\$203,28	\$36,00	R\$609,84	\$108,00
13	Cables	Various cables		Local electronics store	1	R\$112,93	\$20,00	R\$112,93	\$20,00

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Brazilian Real	R\$0,00	\$0,00

#	Name	Software's Tool/Library	Description	Source	Autor	Currency)	Dollars)
1	VL53L0X-Arduino	Library	For controlling the VL53L0X distance sensor on Arduino	pololu/vl53l0x-arduino: Biblioteca	Pololu	FREE	FREE
2	Open CV	Library	Image processing library	OpenCV - Biblioteca abierta de C	Opencv.org	FREE	FREE
3	IDE Arduino	Software's Tool	Software	Software Arduino	Arduino.cc	FREE	FREE
4	Visual Studio Code	Software's Tool	Software	Visual Studio Code: edición de có	Microsoft	FREE	FREE
5	Numpy	Library	Array and math library	NumPy	Numpy developers	FREE	FREE
6	Matplotlib	Library	Python graphics generation library	Matplotlib — Visualización con Py	Matplot lib dev team	FREE	FREE
7	Tinckercad	Software's Tool	Online platform for simulating circuits	Tinkercad	Autodesk	FREE	FREE
8	Git	Software's Tool	Version control for managing functions	Git - Descargas	Software Freedom Conservancy	FREE	FREE
9	Fusion 360	Software's Tool	3D CAD shell design	Fusión de Autodesk Software ba	Autodesk	FREE/EDUCATION	FREE/EDUCATION
10	Tic Arduino	Library	Library for controlling a Nema 17 stepper motor with the Tic T825 driver	https://github.com/pololu/tic-arduir	DavidEGrayson	FREE	FREE
11	Modulemotorl298n	Library	module without having to worry about the pins manually in each sketch.		Name of team members	FREE	FREE

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Euro EUR	R\$507,51	\$558,26

EUR/USD Rate (fixed)
1,1

#	Component	Part name	Author	Source	Quantity	(Local	Unit cost (US Dollars)	(Local	(USA Dollars)
1	Custom PCB	(JLCPCB manufactured)	Goasbock / JLCPCB	m	1	R\$25,00	\$27,50	R\$25,00	\$27,50
2	Microcontroller	(ABX00032)	Arduino	e	1	R\$24,71	\$27,18	R\$24,71	\$27,18
3	Microcontroller	(ABX00083)	Arduino	e	1	R\$20,67	\$22,74	R\$20,67	\$22,74
4	Voltage Regulator	LDO Regulator L4940V5 5V 1.5A (TO-220)	STMicroelectronics	om	2	R\$1,81	\$1,99	R\$3,62	\$3,98
5	Heatsink	Kit	VooGenzek	de	2	R\$0,50	\$0,55	R\$1,00	\$1,10
6	Protection Fuse	3413.0331.22	Schurter	om	1	R\$2,50	\$2,75	R\$2,50	\$2,75
7	Protection Fuse	2920L250DR	Littelfuse	om	1	R\$0,60	\$0,66	R\$0,60	\$0,66
8	I/O Expander IC	(TSSOP-24)	MaxLinear	om	1	R\$1,68	\$1,85	R\$1,68	\$1,85
9	Level Shifter IC	Level Translator (TSSOP-14)	Texas Instruments	om	1	R\$1,31	\$1,44	R\$1,31	\$1,44
10	Level Shifter Module	↔ 5V (Towviy)	Towviy	de	1	R\$2,00	\$2,20	R\$2,00	\$2,20
11	Voltage Monitor	Green Display	Generic	de	1	R\$2,00	\$2,20	R\$2,00	\$2,20
12	Audio Indicator	Buzzer CEM-1206S 92dB 2400Hz 5V THT	Reichelt	e	1	R\$1,26	\$1,39	R\$1,26	\$1,39
13	Switch	3-position THT	C&K	om	1	R\$0,84	\$0,92	R\$0,84	\$0,92
14	Switch	Tactile Push Button 6×6×5mm DIP-2 (×6)	Youmile	de	6	0,05	0,055	R\$0,30	\$0,33
15	Power Connector	(J1/J2)	YIXISI	de	2	0,4	0,44	R\$0,80	\$0,88
16	PCB Connectors	Assortment (J10–J28)	Reichelt / School Stock	e	1	5	5,5	R\$5,00	\$5,50
17	LED Indicator	RGB LED 5mm 4-pin Common Cathode	Youmile	de	1	0,1	0,11	R\$0,10	\$0,11
18	LED Indicator	Red LED 5mm 2V 40000mcd	Lumetheus	de	1	0,05	0,055	R\$0,05	\$0,06
19	Passive Components	Assortment (R1–R26, C1–C26, D1, D5)	Reichelt	e	1	2	2,2	R\$2,00	\$2,20
20	ToF Distance Sensor	Sensor Carrier (×4)	Microelectronics	shop.de	4	23,95	26,345	R\$95,80	\$105,38
21	Color Sensor	Module (×2)	Geekcreit	d.com	2	3,66	4,026	R\$7,32	\$8,05
22	Line Sensor	13582 (I2C)	SparkFun	semaf.at	1	36,35	39,985	R\$36,35	\$39,99
23	Ultrasonic Sensor	HC-SR04 Ultrasonic Distance Sensor	Generic	de	1	3	3,3	R\$3,00	\$3,30
24	Vision System	Pixy2 Smart Camera with SoC (SPI/I2C)	Charmed Labs	com	1	60	66	R\$60,00	\$66,00
25	Motor Driver	(×4)	Texas Instruments	de	4	3	3,3	R\$12,00	\$13,20
26	Drive Motor	100RPM (×4 main drive)	DFRobot	om	4	19,9	21,89	R\$79,60	\$87,56
27	Servo Motor	claw/lift/unload)	Tower Pro	de	3	4	4,4	R\$12,00	\$13,20
28	Display	SPI LCD 2.4" ILI9341 240×320 Module	Generic	de	1	8	8,8	R\$8,00	\$8,80
29	Drive Wheels	compatible Hub (×4)	Generic	de	4	15	16,5	R\$60,00	\$66,00
30	Robot Structure	storage, arm, wheel couplings	Goasbock	Lab P1S	1	8	8,8	R\$8,00	\$8,80
31	Power Supply	connector)	Generic	de	1	25	27,5	R\$25,00	\$27,50
32	Mounting Hardware	PCB Spacers (assortment)	Various	de	1	5	5,5	R\$5,00	\$5,50

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Euro	R\$0,00	\$0,00

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1	C/C++	Programming Language	Hochoptimierte Firmware-Programmiersprache	gcc.gnu.org / arduino.cc	Brüggemann Johannes	FREE	FREE
2	Arduino IDE	Software Tool	Entwicklungsumgebung (IDE) für Arduino	arduino.cc	Arduino (arduino.cc)	FREE	FREE
3	PlatformIO	Software Tool	Erweiterte IDE-Plattform für ESP32-S3 und mehr	platformio.org	Johannes	FREE	FREE
4	PTC Creo	CAD Software	Parametrisches Master-Skeleton-Modelling	ptc.com	Kienberger Markus	EDU FREE	EDU FREE
5	KiCad	Software Tool	Open-Source EDA-Tool für PCB-Schaltpläne	kicad.org	Kienberger Markus	FREE	FREE
6	Git	Software Tool	Versionskontrolle für kollaborative Firmware-Entwicklung	git-scm.com	Brüggemann Johannes	FREE	FREE
7	GitHub	Software Tool	Remote-Repository-Hosting und Team-Projektverwaltung	github.com	Johannes	FREE	FREE
8	HSV Color Space Transf.	Algorithm (Custom)	Konvertierung von RGB-Kamerabild in HSV	Custom implementation	May Roman	FREE	FREE
9	Color Segmentation Algo.	Algorithm (Custom)	Pixelbasierte Segmentierung zur Erkennung von Farben	Custom implementation	May Roman	FREE	FREE
10	Stochastic Object Recognition	Algorithm (Custom)	Statistische Mustererkennung für zuverlässige Objekterkennung	Custom implementation	May Roman	FREE	FREE
11	PID Controller	Algorithm (Custom)	Proportional-Integral-Derivative Regelkreis	Custom implementation	Johannes	FREE	FREE
12	Interrupt-Driven RTOS	Firmware Architecture	Echtzeit-Firmware mit Interrupt-gesteuerten Tasks	Custom implementation	Johannes	FREE	FREE
13	Line Following State Machine	Algorithm (Custom)	Zustandsautomat für Linienverfolgung basierend auf Sensordaten	Custom implementation	Johannes	FREE	FREE
14	Obstacle Avoidance Logic	Algorithm (Custom)	Hinderniserkennung und Ausweichstrategie	Custom implementation	Johannes	FREE	FREE
15	Intersection Logic	Algorithm (Custom)	Kreuzungserkennungs- und Entscheidungslogik	Custom implementation	Brüggemann Johannes	FREE	FREE
16	Rescue Zone Algorithm	Algorithm (Custom)	Automatische Erkennung der Rescue Zone	Custom implementation	May Roman	FREE	FREE
17	Pixy2 Arduino Library	Library	Kommunikationsbibliothek für Pixy2 Kamera	pixycam.com / github.com/charmedlabs/pixy2	May Roman	FREE	FREE
18	VL53L5CX Arduino Library	Library	ST-Treiber für Time-of-Flight Sensor VL53L5CX	st.com / github.com/stmicroelectronics	Brüggemann Johannes	FREE	FREE
19	SparkFun Line Follower Lib.	Library	Bibliothek für SparkFun IR Line Follower Arduino	sparkfun.com / github.com/sparkfun	Brüggemann Johannes	FREE	FREE
20	Wire Library (I2C)	Library	Arduino I2C-Kommunikationsbibliothek für Wire	arduino.cc	Brüggemann Johannes	FREE	FREE
21	SPI Library	Library	Arduino SPI-Bibliothek für ILI9341 LCD Display	arduino.cc	Brüggemann Johannes	FREE	FREE
22	Adafruit ILI9341 Library	Library	Grafik-Bibliothek für 2.4 Zoll SPI LCD Display	adafruit.com / github.com/adafruit	Kienberger Markus	FREE	FREE